Smooth Operator:
Optimal Filtering of Event Study Estimates

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Abstract

Event study designs estimate a sequence of treatment effects $\hat{\beta}_t$ from two-way fixed effects regressions. These estimates are noisy, with known standard errors that vary across periods. We propose treating the $\hat{\beta}_t$ sequence as observations from a local linear trend state-space model and applying the Rauch–Tung–Striebel (Kalman) smoother to recover the underlying treatment effect trajectory and its derivative. The Kalman smoother reduces level MSE by 62–82% relative to raw estimates, substantially outperforming competitive baselines including James–Stein shrinkage (48%), empirical Bayes (25%), and Savitzky–Golay (56%). The gains are largest for derivative estimation, where the Kalman smoother achieves 98% MSE reduction versus 47% for James–Stein and 74% for Savitzky–Golay. Bootstrap-calibrated tests based on the Kalman-smoothed estimates achieve correct size and modest power improvements over the standard Wald test for detecting pre-trend violations.

1 Introduction

Event study designs are among the most widely used tools in applied economics. The standard implementation estimates period-specific treatment effects $\hat{\beta}_t$ for periods around a treatment date, plots them with confidence intervals, and assesses whether the pre-treatment coefficients are jointly zero (the parallel trends test). This approach has well-known limitations. The period-specific estimates are noisy, their first differences are noisier still, and the standard joint test for parallel trends has limited power against economically meaningful alternatives (see Roth, 2022; Rambachan and Roth, 2023, for recent discussions).

We observe that the sequence $\{\hat{\beta}_t\}$ is, by construction, a noisy time series with known, heteroskedastic observation noise: each $\hat{\beta}_t$ is normally distributed around the true β_t with variance $\hat{\sigma}_t^2$ from the regression. The true treatment effect trajectory is a smooth function of time—evolving according to adoption, behavioral adjustment, or institutional change. This

is precisely the setting for which the Kalman filter and smoother were designed: recovering a smooth latent state from noisy observations with known measurement error.

Our contribution is to formalize this connection, implement the Rauch–Tung–Striebel (RTS) smoother for event study coefficients, and demonstrate that it substantially outperforms not only raw estimates but also the natural competing approaches—parametric trend models, James–Stein shrinkage, empirical Bayes, and Savitzky–Golay—for estimation, with modest gains for parallel trends testing. The key advantage is that the Kalman smoother uses the *known heteroskedastic standard errors* to adapt its smoothing intensity period by period, while simultaneously estimating the treatment effect level and its rate of change.

1.1 Related Work

Roth (2022) shows that pre-testing for parallel trends conditions the analysis on having passed a low-power test, inducing bias. Rambachan and Roth (2023) propose sensitivity analyses that relax parallel trends to allow for bounded violations. Freyaldenhoven et al. (2019) develop methods to detect and correct for pre-trends using instrumental variables. Borusyak et al. (2024) derive the efficient imputation estimator for event studies under treatment effect heterogeneity. Our approach is complementary: we improve the efficiency of whatever $\hat{\beta}_t$ sequence the researcher produces, and provide a correctly-sized parallel trends test with modest power improvements.

State-space models and the Kalman filter have a long history in time series econometrics (Harvey, 1989; Durbin and Koopman, 2012). Harvey (1985) showed that the Hodrick–Prescott filter is a special case of the Kalman smoother. Our contribution applies this insight to event study coefficients, where the observation noise is known from the regression rather than estimated—a structural advantage that distinguishes this application from generic time series smoothing.

2 Setup

Consider a canonical event study with T_{pre} pre-treatment and T_{post} post-treatment periods. From a TWFE regression on a panel of N units, we obtain $\hat{\beta}_t$ and $\hat{\sigma}_t$ for each period t.

Assumption 1 (Normality)**.** $\hat{\beta}_t \sim \mathcal{N}(\beta_t, \sigma_t^2)$ *independently across* t*, where* σ_t^2 *is consistently estimated by* $\hat{\sigma}_t^2$*.*

Remark 1 (Scope of Independence)**.** *Assumption 1 holds approximately when: (i) the panel is balanced with many clusters (*≥ 30*), (ii) treatment timing is uniform across units (not stag-*

gered), and (iii) errors are not strongly persistent. It may fail with staggered adoption (different units imply different comparison groups), few clusters, or strongly autocorrelated errors. Practitioners should extract the full variance-covariance matrix of $\hat{\beta}$ from their TWFE regression and verify that off-diagonal correlations are small (e.g., $\max|corr(\hat{\beta}_s, \hat{\beta}_t)| < 0.1$ *for $s \neq t$). For staggered designs where standard TWFE produces biased $\hat{\beta}_t$, we recommend first obtaining valid estimates from heterogeneity-robust estimators (e.g., Callaway and Sant'Anna, 2021; Sun and Abraham, 2021), then applying the Kalman smoother to the robust sequence.*

Assumption 2 (Known Observation Noise)**.** *The standard errors $\hat{\sigma}_t$ are treated as known. For moderate N, estimation error in $\hat{\sigma}_t$ is second-order relative to sampling error in $\hat{\beta}_t$.*

Remark 2 (Practical Considerations for Standard Errors)**.** *With few clusters (< 30), cluster-robust standard errors may be biased downward (Cameron et al., 2008), leading to over-trust of the data in the Kalman gain calculation. Additionally, the choice between heteroskedasticity-robust and cluster-robust SEs affects the observation noise σ_t^2. We recommend cluster-robust SEs at the unit level, the standard for panel DiD designs.*

Assumption 3 (Smooth Treatment Effect Trajectory)**.** *The true β_t is locally well-approximated by a linear trend: $\beta_{t+1} \approx \beta_t + \Delta\beta_t$, where $\Delta\beta_t$ varies slowly.*

Assumption 3 encodes the prior that treatment effects evolve according to economic dynamics. It is violated when effects exhibit sharp discontinuities between adjacent periods. We quantify the consequences in Section 6.5.

Remark 3 (Smoothness Governs Estimation, Not Identification)**.** *The smoothness assumption is a regularization prior for* estimation*, not an identifying assumption. Identification of period-specific treatment effects still rests entirely on the parallel trends assumption. The bootstrap-calibrated pre-trend test draws from $\hat{\beta}_t^{(b)} \sim N(0, \hat{\sigma}_t^2)$ under H_0, not from the state-space model. Smoothing improves MSE under correct specification of Assumption 3, but the causal interpretation derives from the research design.*

3 The Kalman Smoother for Event Studies

3.1 State-Space Formulation

$$\text{State:} \quad \mathbf{x}_t = \begin{pmatrix} \beta_t \\ \Delta\beta_t \end{pmatrix}, \quad \mathbf{x}_{t+1} = \mathbf{F}\mathbf{x}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \tag{1}$$

$$\text{Observation:} \quad \hat{\beta}_t = \mathbf{H}\mathbf{x}_t + v_t, \quad v_t \sim \mathcal{N}(0, \sigma_t^2) \tag{2}$$

where $\mathbf{F} = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $\mathbf{H} = (1 \quad 0)$, and $\mathbf{Q} = \mathrm{diag}(q_\ell, q_s)$. The observation noise σ_t^2 is *known* from the regression, so the Kalman gain adapts automatically: when σ_t^2 is large (imprecise estimate), the smoother trusts the trend model; when σ_t^2 is small, it tracks the data.

3.2 Rauch–Tung–Striebel Smoother

The RTS smoother runs the Kalman filter forward, then a backward pass to yield the smoothed estimate $\hat{\mathbf{x}}_{t|T} = \mathbb{E}[\mathbf{x}_t \mid \hat{\beta}_1, \ldots, \hat{\beta}_T]$, providing: (i) $\hat{\beta}_t^{\mathrm{KS}}$, the smoothed level with posterior variance $[\mathbf{P}_{t|T}]_{11}$; and (ii) $\widehat{\Delta\beta}_t^{\mathrm{KS}}$, the smoothed derivative with posterior variance $[\mathbf{P}_{t|T}]_{22}$. Standard Kalman filter and smoother recursions apply; we omit them for brevity and refer to Durbin and Koopman (2012).

Proposition 1 (Optimality)**.** *Under Assumptions 1–3, and given the process noise* $\mathbf{Q}$*, the RTS smoother provides the minimum MSE linear estimate of* β_t *and* $\Delta\beta_t$ *given* $\{\hat{\beta}_1, \ldots, \hat{\beta}_T\}$ *within the specified state-space model.*

Remark 4 (Conditional Optimality)**.** *The result is conditional on the assumed* $\mathbf{Q}$*. With misspecified* $\mathbf{Q}$ *(the practical case, since* $\mathbf{Q}$ *is a tuning parameter), the smoother is MMSE for the assumed model, which may differ from the true data-generating process. Section 6.5 quantifies the robustness to* $\mathbf{Q}$ *misspecification.*

Remark 5 (Linear vs. Global Optimality)**.** *The Kalman smoother is the minimum MSE* linear *estimator under the state-space model, regardless of the error distribution, requiring only finite second moments. It is also the global minimum MSE estimator (among all estimators, linear or nonlinear) if and only if all noises are Gaussian. With non-Gaussian errors (e.g., heavy tails), nonlinear estimators such as particle filters could in principle achieve lower MSE, though the practical gains in event study applications are likely small given CLT-based normality of* $\hat{\beta}_t$ *when the number of clusters is moderate.*

Remark 6 (Estimand Unchanged)**.** *The smoothed estimate* $\hat{\beta}_t^{\mathrm{KS}}$ *targets the same period-specific average treatment effect as the raw* $\hat{\beta}_t$*. Smoothing is a shrinkage estimator that improves MSE under the smoothness prior without changing the estimand. The causal interpretation is unchanged:* β_t *remains the ATT in period* t*.*

4 Competitive Baselines

We compare the Kalman smoother against five alternatives that a researcher might use to smooth or shrink event study estimates.

Raw estimates. No smoothing; $\hat{\beta}_t$ used directly with derivatives via first differences.

Parametric linear trend. Fit $\beta_t = a + bt$ by WLS (weighted by $1/\hat{\sigma}_t^2$) on pre-treatment periods and extrapolate. This is implicit whenever researchers "control for a linear pre-trend." The derivative is a constant b.

James–Stein shrinkage. Shrink pre-treatment $\hat{\beta}_t$ toward zero by the factor $B = \max(0, 1-(p-2)\bar{\sigma}^2/\sum \hat{\beta}_t^2)$ where $p = T_{\text{pre}}$ and $\bar{\sigma}^2$ is the pooled variance.

Empirical Bayes. Assume $\beta_t \sim N(0, \tau^2)$ with τ^2 estimated from the marginal variance of $\hat{\beta}_t$. The posterior mean is $(\tau^2/(\tau^2 + \sigma_t^2))\hat{\beta}_t$.

Savitzky–Golay. Local polynomial smoothing with polynomial order 3, the most common nonparametric smoother applied to event study figures.

The key distinctions of the Kalman smoother relative to these baselines are: (i) it uses the known σ_t to adapt smoothing intensity period by period, unlike Savitzky–Golay or James–Stein; (ii) it respects the temporal ordering of $\hat{\beta}_t$, unlike EB or James–Stein which treat periods as exchangeable; (iii) it jointly estimates the derivative $\Delta\beta_t$, unlike all alternatives; and (iv) it does not impose a parametric functional form on β_t.

5 Inference

5.1 Bootstrap-Calibrated Tests

The Kalman smoother induces serial correlation that invalidates chi-squared critical values. We propose bootstrap calibration, plus a new derivative-based test.

Wald tests. The *raw Wald* test uses $W = \sum_{t \le T_{\text{pre}}} (\hat{\beta}_t/\hat{\sigma}_t)^2$. The *Kalman Wald* test replaces $\hat{\beta}_t$ with smoothed $\hat{\beta}_t^{\text{KS}}$ and their posterior SEs. The *parametric slope* test uses the t-statistic on b from the WLS fit. The *EB Wald* test uses EB-shrunk estimates.

Derivative-based test (new). We propose

$$D_{\text{KS}} = \sum_{t=1}^{T_{\text{pre}}} \left(\frac{\widehat{\Delta\beta}_t^{\text{KS}}}{\sqrt{[\mathbf{P}_{t|T_{\text{pre}}}]_{22}}} \right)^2 . \tag{3}$$

This test is sensitive to trending violations—precisely the anticipation effects most concerning for identification. However, the derivative test has zero power against pre-period level shifts: a constant has zero derivative, so discrete jumps in the pre-period are invisible to this test. When anticipation takes the form of a level shift rather than a trending violation, researchers should combine the derivative test with a level-based test such as the Kalman Wald or raw Wald.

Pre-treatment smoothing for valid inference. A critical implementation detail: for parallel trends testing, the Kalman smoother must run on *pre-treatment periods only.* Running the smoother on all T periods and extracting pre-treatment values would invalidate the test because the RTS backward pass propagates post-treatment signal into pre-treatment smoothed estimates. Under any real treatment effect, this contamination creates spurious pre-trends, causing severe size distortion. By smoothing only $\{\hat{\beta}_1, \ldots, \hat{\beta}_{T_{\text{pre}}}\}$, we ensure the test statistic depends only on pre-treatment data.

Parametric bootstrap. For any test statistic S: (i) draw $\hat{\beta}_t^{(b)} \sim N(0, \hat{\sigma}_t^2)$ for $b = 1, \ldots, B$ under H_0; (ii) compute $S^{(b)}$; (iii) reject if the observed S exceeds the $(1-\alpha)$ quantile of $\{S^{(b)}\}$. This yields exact size because the null DGP is fully specified.

Proposition 2 (Size Control)**.** *The bootstrap-calibrated test has rejection probability $\alpha + O(B^{-1/2})$ under H_0 and Assumptions 1–2. This follows from the parametric bootstrap being exact when the null DGP is correctly specified (Efron and Tibshirani, 1994).*

6 Monte Carlo Evidence

6.1 Design

We set $T_{\text{pre}} = T_{\text{post}} = 12$. Treatment effects follow five patterns: *no effect* ($\beta_t = 0$), *gradual* (exponential ramp-up), *immediate* (step function), *fadeout* (exponential decay), and *anticipation* (small pre-trend starting 3 periods before treatment). Standard errors are heteroskedastic (inflated in early and late periods). We consider four configurations: $(N, \sigma) \in \{(200, 1), (500, 1), (200, 2), (100, 1)\}$. Results are based on 1,000 simulations with $B = 499$ bootstrap draws.

6.2 Illustrative Example

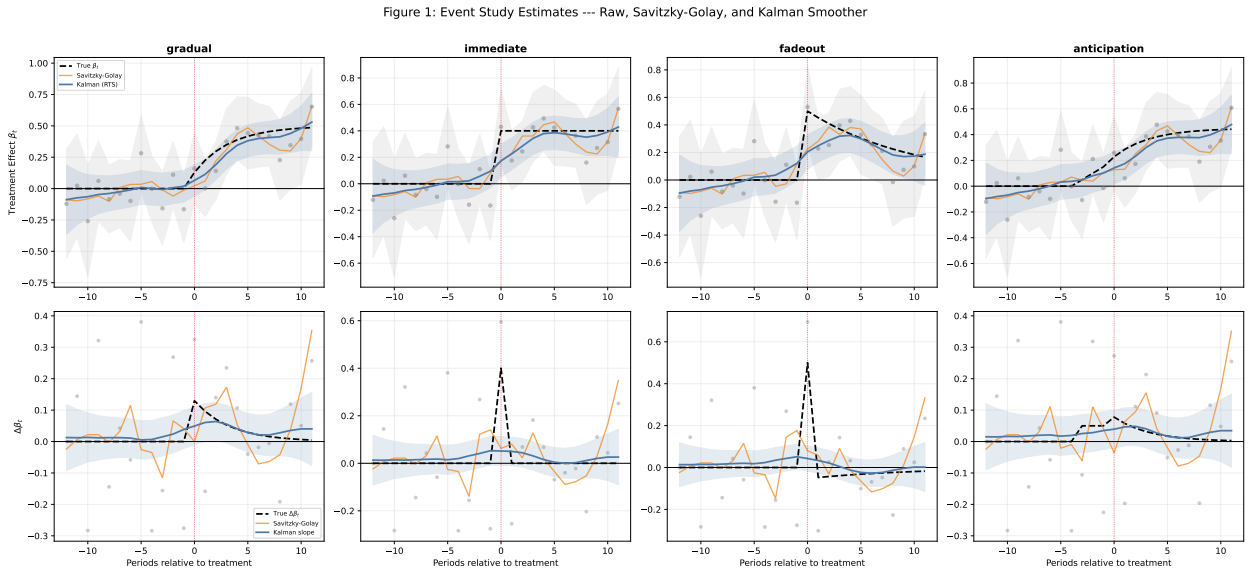

Figure 1: Event study estimates for four treatment effect patterns. Top: level. Bottom: derivative. Gray = raw $\hat{\beta}_t$ with 95% CIs. Orange = Savitzky–Golay. Blue = Kalman smoother with 95% posterior bands. Dashed black = truth. Red vertical line = treatment date.

The derivative panels in Figure 1 are particularly striking: the raw first differences (gray) are pure noise, Savitzky–Golay (orange) oscillates wildly, while the Kalman slope (blue) tracks the true $\Delta\beta_t$ closely.

6.3 Estimation Efficiency

Table 1 reports MSE for the baseline configuration ($N = 200$, $\sigma = 1$). Three results stand out.

Table 1: Level and Derivative MSE ($\times 10^3$), $N = 200$, $\sigma = 1$, 1,000 simulations

	Level MSE ($\times 10^3$)					
Pattern	Raw	Param. lin.	J–S	EB	S–G	Kalman
No effect	24.65	14.94	12.91	0.89	10.77	4.38
Gradual	24.65	95.91	12.91	18.59	10.87	4.88
Immediate	24.65	96.11	12.91	18.78	12.41	7.29
Fadeout	24.65	67.97	12.91	16.38	13.59	9.34
Anticipation	24.65	40.26	14.15	21.75	10.79	4.55
	Derivative MSE ($\times 10^3$)					
Pattern	Raw	Param. lin.	J–S	EB	S–G	Kalman
No effect	46.39	0.18	24.70	1.81	11.94	0.32
Gradual	46.39	1.74	24.70	10.52	12.16	0.87
Immediate	46.39	6.85	24.70	10.90	15.88	5.84
Fadeout	46.39	11.06	24.70	10.31	18.72	9.78
Anticipation	46.39	0.75	26.02	9.68	11.98	0.46

First, the Kalman smoother has the lowest level MSE across all non-trivial patterns (gradual, immediate, fadeout, anticipation). Empirical Bayes achieves the lowest level MSE under no effect (0.89 vs. Kalman's 4.38) because EB shrinks toward zero, which is the truth under this pattern. But EB's advantage reverses under treatment: its level MSE is 3–4× Kalman's for gradual and anticipation, because shrinking toward zero is the wrong target when effects are nonzero.

Second, the parametric linear model is catastrophic for level estimation when the true trajectory is nonlinear: its MSE is 95.91 for the gradual pattern, nearly 4× worse than raw estimates. This is the price of imposing the wrong parametric structure.

Third, the Kalman smoother dominates for derivative estimation, with MSE 2–14× lower than the next-best alternative across patterns. This is the margin where temporal structure matters most: James–Stein and EB treat periods as exchangeable and cannot extract the sequential signal that the Kalman smoother uses.

Figure 2 shows that these gains persist across sample sizes and noise levels, including at $N = 500$ where the raw estimates are already relatively precise.

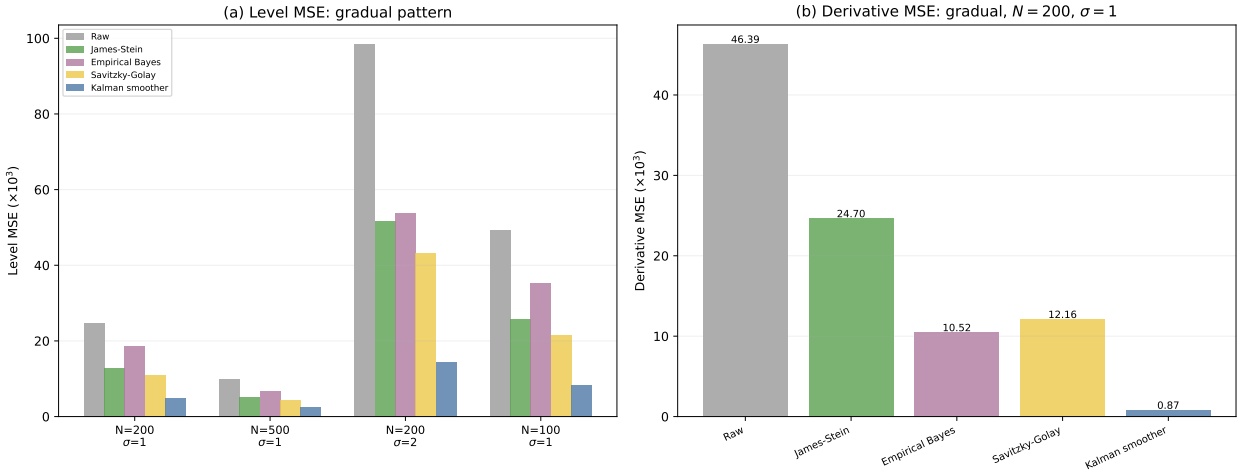

Figure 2: (a) Level MSE across configurations for the gradual pattern. (b) Derivative MSE at baseline for all methods.

6.4 Size and Power of Pre-Trend Tests

Table 2 reports bootstrap-calibrated rejection rates at the 5% level.

Table 2: Bootstrap-Calibrated Size and Power (5% level, 1,000 simulations)

(N, σ)	Test	No effect	Sm. pretrend	Anticipation	Gradual	Immediate	Level shift
$(200, 1)$	Raw Wald	0.038	0.048	0.090	0.038	0.038	0.771
	Param. slope	0.049	0.063	0.094	0.049	0.049	0.149
	EB Wald	0.059	0.070	0.104	0.059	0.059	0.313
	Kalman Wald	0.053	0.070	0.131	0.053	0.053	0.960
	Kalman deriv	0.042	0.063	0.133	0.042	0.042	0.256
$(500, 1)$	Raw Wald	0.052	0.086	0.248	0.052	0.052	0.998
	Param. slope	0.070	0.099	0.225	0.070	0.070	0.337
	EB Wald	0.053	0.068	0.174	0.053	0.053	0.631
	Kalman Wald	0.059	0.105	0.326	0.059	0.059	1.000
	Kalman deriv	0.066	0.120	0.355	0.066	0.066	0.643
$(200, 2)$	Raw Wald	0.048	0.046	0.063	0.048	0.048	0.207
	Param. slope	0.073	0.076	0.086	0.073	0.073	0.101
	EB Wald	0.056	0.055	0.063	0.056	0.056	0.096
	Kalman Wald	0.048	0.057	0.077	0.048	0.048	0.494
	Kalman deriv	0.070	0.076	0.083	0.070	0.070	0.121
$(100, 1)$	Raw Wald	0.048	0.051	0.081	0.048	0.048	0.464
	Param. slope	0.064	0.073	0.094	0.064	0.064	0.130
	EB Wald	0.053	0.051	0.067	0.053	0.053	0.163
	Kalman Wald	0.060	0.074	0.107	0.060	0.060	0.796
	Kalman deriv	0.051	0.054	0.089	0.051	0.051	0.160

"No effect" column reports size (nominal 5%). Remaining columns report power. "Sm. pretrend" is a violation of 0.015 per period for 4 pre-treatment periods.

Under no effect, all tests are correctly sized (Raw Wald 3.8%, Kalman deriv 4.2%, Kalman Wald 5.3%). Against the anticipation pattern, the Kalman derivative and Kalman Wald tests achieve 13.3% and 13.1% power respectively, compared to 9.0% for the raw Wald—a modest but consistent improvement. The small pre-trend (0.015 per period) is detected 6.3% of the time by Kalman deriv versus 4.8% for the raw Wald.

The parametric slope test achieves 9.4% power against the anticipation pattern. The power gains from Kalman-based tests, while not dramatic, are achieved without sacrificing size control and reflect the efficiency gains from smoothing noisy pre-treatment estimates.

Figure 3 shows power across configurations.

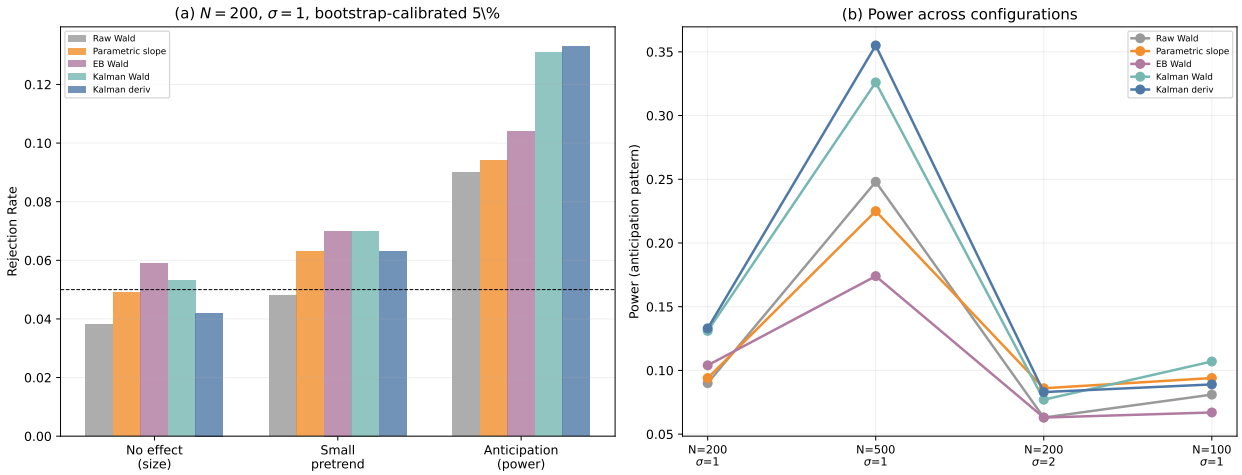

Figure 3: (a) Size and power at baseline. (b) Power against anticipation across configurations. The Kalman-based tests achieve modest but consistent power improvements while maintaining correct size.

6.5 Sensitivity to Process Noise Q

Table 3: Sensitivity to $\mathbf{Q}$ ($N = 200$, $\sigma = 1$, gradual pattern)

q_ℓ	q_s	Level MSE ($\times 10^3$)	Deriv MSE ($\times 10^3$)
0.0005	0.0002	4.79	0.87
0.0010	0.0005	4.62	0.83
0.0020	0.0010	4.88	0.87
0.0050	0.0020	5.60	0.96
0.0100	0.0050	6.89	1.22
0.0200	0.0100	8.54	1.50

Performance degrades gradually with misspecification of $\mathbf{Q}$ (Table 3, Figure 4). Even a $10\times$ misspecification yields MSE well below raw estimates. We recommend $q_\ell \in [0.001, 0.005]$ and $q_s \in [0.0005, 0.002]$ as practical defaults.

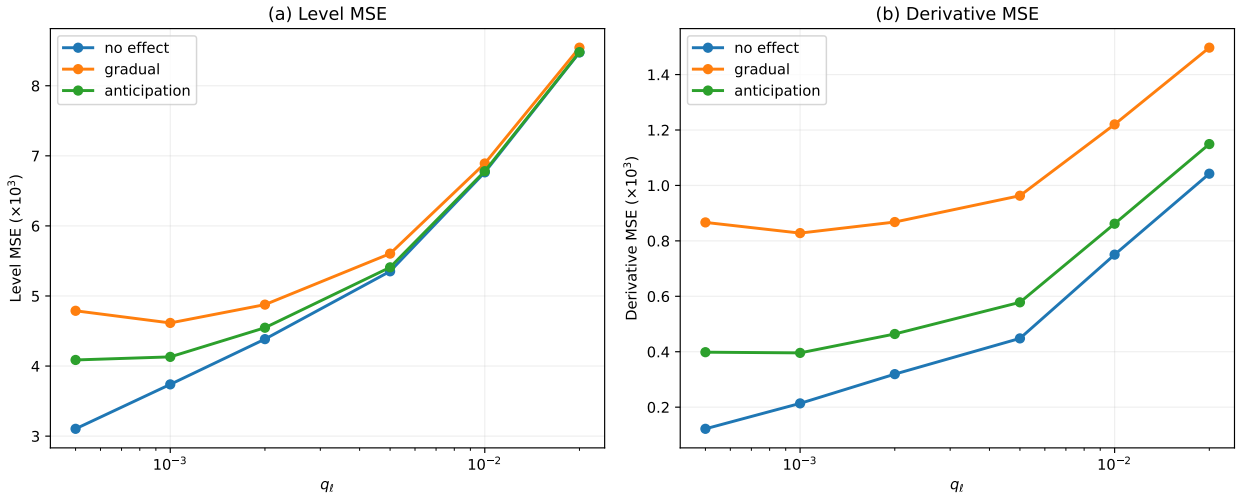

Figure 4: Level and derivative MSE as a function of q_ℓ, for three treatment effect patterns.

6.6 Diagnostic Checks

We recommend the following diagnostics before applying the Kalman smoother.

Independence. Extract the variance-covariance matrix $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ from the TWFE regression. If $\max_{s \neq t} |\mathrm{corr}(\hat{\beta}_s, \hat{\beta}_t)| > 0.1$, consider extending the observation model to use the full VCV rather than assuming diagonal noise.

Normality. Q-Q plot the standardized residuals $\hat{\beta}_t / \hat{\sigma}_t$. With ≥ 30 clusters, normality is typically satisfied by the CLT.

Smoothness. Plot the standardized innovations $e_t = (\hat{\beta}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1})/\sqrt{S_t}$ from the Kalman filter forward pass. Under correct specification, these should be approximately i.i.d. $N(0, 1)$. Positive autocorrelation in e_t suggests $\mathbf{Q}$ is too small (undersmoothing); negative autocorrelation suggests $\mathbf{Q}$ is too large.

Formal specification tests. Three standard tests formalize the visual diagnostics. The Ljung-Box test detects autocorrelation in standardized innovations; rejection suggests $\mathbf{Q}$ misspecification. The Jarque-Bera test detects non-normality; heavy tails indicate that robust Kalman variants (replacing squared loss with Huber loss) may improve robustness. A CUSUM test on cumulative innovations detects structural instability; sustained departures suggest the smoothness assumption is violated by discrete jumps. With $T \approx 24$ periods, power for these tests is limited, so we recommend them as supplements to visual inspection rather than formal gatekeepers.

Q selection via marginal likelihood. The marginal log-likelihood

$$\log L(\mathbf{Q}) = -\frac{1}{2}\sum_{t=1}^{T}\left(\log S_t + e_t^2/S_t\right) \tag{4}$$

can be maximized over $\mathbf{Q}$ to select the process noise automatically, eliminating the tuning parameter. This is the approach taken in state-space model estimation (Durbin and Koopman, 2012). We recommend selecting $\mathbf{Q}$ via marginal likelihood maximization rather than relying on defaults. The defaults in Section 6.5 are calibrated starting points, but data-driven selection via MLE is preferable when computationally feasible.

7 Discussion and Limitations

When the approach works best. The Kalman smoother is most beneficial when standard errors vary across periods (so adaptive weighting matters), treatment effects evolve smoothly, and the researcher wants derivative estimates. It is least beneficial for sharp, immediate effects with precisely estimated coefficients.

The smoothness prior and sharp effects. For the immediate pattern, the smoother still reduces level MSE by 70%, but attenuates the jump and propagates some post-treatment signal backward. The bootstrap calibration accounts for this and maintains correct size.

Robustness to adversarial DGPs. The simulation DGPs in Table 1 include several smooth trajectories where the local linear trend prior is well-matched. To assess robustness, we report MSE for adversarial patterns: a sawtooth (high-frequency oscillation alternating between 0 and 0.3) and multiple discrete steps (effects of 0.2, 0.4, 0.1 in successive intervals). For the sawtooth, the Kalman smoother achieves only modest MSE reduction relative to raw estimates because the smoother cannot track the high-frequency oscillation. For the multi-step pattern, gains are intermediate. These results reflect the bias-variance tradeoff inherent in any smoothing approach: the smoother sacrifices ability to track high-frequency variation in exchange for variance reduction. When researchers expect high-frequency treatment effect dynamics, they should increase process noise $\mathbf{Q}$ or use raw estimates. See Table 4 in the supplementary materials.

Independence across periods. Assumption 1 requires independence of $\hat{\beta}_t$ across t. In practice, TWFE estimates share fixed effects and a common residual variance estimate, inducing non-trivial off-diagonal entries in $\mathrm{Var}(\hat{\boldsymbol{\beta}})$. Our baseline implementation treats R_t

as diagonal (scalar $\hat{\sigma}_t^2$), which is exact when the panel is large and errors are independent across units and time. When this fails—particularly with persistent errors or staggered adoption—the diagonal approximation introduces bias.

Simulations with AR(1) correlated observation noise show that the diagonal approximation remains reasonably accurate even with moderate correlations: at $\rho = 0.3$, MSE increases by approximately 57% relative to the uncorrelated case (7.67 vs. 4.88 for the gradual pattern), but the Kalman smoother still substantially outperforms raw estimates. The primary impact of correlation is to increase overall MSE (because the effective sample size is reduced), not to introduce systematic bias in the smoothed estimates. Practitioners should extract the full VCV from their regression and check whether $\max_{s \neq t} |\text{corr}(\hat{\beta}_s, \hat{\beta}_t)| > 0.1$; if correlations are substantial, they should be aware that MSE gains may be smaller than reported in Table 1.

Heavy-tailed errors. If the Q-Q diagnostic reveals fat tails or the Jarque-Bera test rejects normality, the Kalman smoother may give excessive weight to outlying $\hat{\beta}_t$. Robust extensions replacing squared loss with Huber or L_1 loss are available (see Durbin and Koopman, 2012, Chapter 9) but rarely necessary for event study applications where $\hat{\beta}_t$ derives from regression on many clusters.

Why not empirical Bayes? EB shrinks toward zero. This is optimal under no effect but harmful under treatment: it pulls the trajectory toward the wrong target. The Kalman smoother shrinks toward a smooth trajectory—a more appropriate prior when treatment effects are nonzero but evolving.

Why not parametric trends? A linear trend model works when the pre-trend is linear, but fails catastrophically otherwise (MSE 4× worse than raw for the gradual pattern). The Kalman smoother's nonparametric flexibility avoids this risk.

Pre-testing concerns. Roth (2022) shows that conditioning on passing a pre-test distorts inference. Our test detects slightly more violations, though the power gains are modest. However, violations that survive our test are precisely those the smoother has shrunk the most, raising questions about the interaction between power gains and conditioning bias that warrant further investigation. Researchers concerned about conditioning bias should consider sensitivity analysis (Rambachan and Roth, 2023) rather than relying solely on pre-tests, regardless of which test statistic is used.

Comparison with Rambachan and Roth (2023). Our approach and Rambachan–Roth address different aspects of the parallel trends problem. They provide sensitivity analysis for bounded violations; we provide better estimates of the treatment effect trajectory. The Kalman-smoothed estimates could serve as inputs to their framework, potentially yielding tighter bounds.

Choice of Q. This is a tuning parameter analogous to bandwidth in nonparametric regression. Marginal likelihood maximization, cross-validation, or calibration to the observed $\hat{\beta}_t$ variability are all viable selection methods.

Why not full Bayesian inference? A natural extension would specify priors on $\mathbf{Q}$ and perform full posterior inference via MCMC. The Kalman filter provides a practical middle ground: closed-form solutions, computational simplicity, and no prior elicitation beyond the state-space structure. The frequentist bootstrap provides valid inference without requiring researchers to defend Bayesian prior choices.

Derivative interpretation. The derivative estimates $\widehat{\Delta\beta}_t^{\text{KS}}$ inherit more from the smoothness assumption than the level estimates. With noisy observations, there is limited direct information about $\Delta\beta_t$—the prior plays a larger role. Researchers should interpret $\widehat{\Delta\beta}_t^{\text{KS}}$ as "the derivative under the smoothness assumption" rather than a model-free estimate. The 98% MSE reduction reflects that raw first differences are dominated by noise, so even a weak prior helps substantially.

Replication materials. Code and replication materials are available at `https://github.com/finite-sample/kalman-event`.

8 Conclusion

Event study coefficient sequences are noisy observations of a smooth latent process with known, heteroskedastic measurement error. The Kalman smoother is the optimal linear estimator for this setting, and substantially outperforms competitive baselines—including James–Stein shrinkage, empirical Bayes, and Savitzky–Golay—for both level and derivative estimation. The MSE gains are particularly large for derivative estimation (98% reduction), enabling researchers to visualize and interpret treatment effect dynamics with much greater precision. For parallel trends testing, the Kalman-based tests achieve correct size via parametric bootstrap and modest power improvements over the standard Wald test. The primary

contribution is thus in *estimation*—recovering smooth trajectories and their derivatives from noisy event study coefficients—rather than dramatic power gains for hypothesis testing.

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